

Football Fan Community Analysis — r/soccer

COSC2671 / COSC3047 — Social Media and Network Analysis

Assignment 2 — Group Project

Team Members

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1.INTRODUCTION

The emergence of social media platforms has provided a new arena for sports fans to engage in real time on matches, players and the tournament proceedings. Big international sports events like the FIFA World Cup generate extremely high levels of interaction and engagement in online communities. The discussions are important opportunities for using social media and network analysis techniques to analyse user behaviour, sentiment and communication patterns.

Reddit is one of the biggest social media sites for discussions and has very active sports communities. One of the most popular online football communities, r/soccer has users from all over the world discussing match reactions, player performances, transfer news, refereeing controversies, and fan opinions.

This project studies discussions on Reddit (r/soccer) during the 2022 FIFA World Cup. In this paper, we use a combination of Natural Language Processing (NLP) and Social Network Analysis (SNA) techniques to analyse interactions among football fans in the web, to find out the most discussed topics and to identify influential users in the community.

The project analyses behavioural patterns in the discussion network using sentiment analysis, topic modelling, centrality analysis and community detection. The project combines text analysis and graph-based analysis and aims to give a wider understanding of the interactions of football fans during a major global sporting event.

2. Research Question and Objectives

Research Question

What do interaction networks, sentiment polarity and discussion topics tell us about the social behaviour and influential communities of Reddit football fandom during the 2022 FIFA world cup?

Objectives

The goals of this project are to:

- Build a Reddit user interaction network based on reply relations.
- Apply network centrality metrics to pinpoint influential users.
- Use graph clustering methods for user community identification.
- Analyse sentiment patterns of football related discussion.
- Topic modelling to identify dominant discussion topics.
- Study relationships between influence, sentiment and interaction behaviour.

Success Criteria

The project is deemed successful if it is able to do:

- Build an interpretable and meaningful interaction network.
- Find influential users in the community.
- Discover coherent user communities..
- Extract meaningful topics of football discussions.
- Analyse sentiment trends over discussions.
- Present evidence-based conclusions supported by the analysis results.

3. Dataset Description

3.1 Data Source

The dataset for this project was obtained from the online forum Reddit, specifically from the subreddit r/soccer. We chose Reddit because of the large scale discussion threads and active football communities that generate high levels of interaction during major sporting events.

The analysis was limited to conversations that took place during the 2022 FIFA World Cup period (November 2022 to December 2022). The reason for this time frame is that the tournament generated a lot of global engagement and discussion activity across football communities.

The dataset includes:

- Reddit posts,
- Reddit comments,
- user interaction data,
- timestamps,
- metadata related to user engagement.

The data was collected using Python-based Reddit data extraction scripts and stored in JSON format for preprocessing and analysis.

3.2 Dataset Structure

Two primary datasets were used:

Posts Dataset

The posts dataset contains submission-level information from r/soccer, including:

- post titles,
- post authors,
- post scores,
- timestamps,

- number of comments,
- post flair categories.

Comments Dataset

The comments dataset contains comment-level discussion data, including:

- comment text,
- comment authors,
- reply relationships,
- comment scores,
- timestamps,
- parent-child interaction information.

The final dataset consisted of approximately:

- 18088 Reddit posts,
- 45,020 Reddit comments,
- thousands of unique users participating in discussions.

3.3 Important Variables

Several fields were used throughout the analysis.

Variable	Description
author	Reddit username
body	Comment text
title	Reddit post title
created_utc	Unix timestamp of post/comment
parent_id	Parent post or parent comment ID
score	Reddit upvote score
subreddit	Source subreddit
flair	Post category or discussion label

The parent_id field was especially important to build the user interaction network. The body and title fields were used for NLP based analysis such as sentiment analysis and topic modelling.

3.4 Data Preprocessing

Before proceeding to analysis, some preprocessing steps were performed:

- Removed deleted users and bot accounts
- Converted timestamps to datetime format
- Removed missing or invalid records
- Cleaned text content, combined post title and body text
- Filtered out empty comments and posts
- Constructed reply relationships between users.

These preprocessing steps improved the quality of the dataset and reduced noise in both the NLP and network analysis components.

3.5 Suitability for Network Analysis

The dataset fitted to network analysis very well since discussions on Reddit naturally have reply relationships between users. These reply interactions enable us to build a directed interaction graph where:

- nodes represent Reddit users,
- edges represent reply interactions between users,
- edge weights represent the frequency of replies.

This structure enabled analysis of:

- user influence,
- interaction behaviour,
- community formation,
- communication patterns.

3.6 Suitability for NLP Analysis

- The data set was also suitable for NLP analysis, because there was a lot of unstructured textual data on discussions of football events and fans' reactions.

The text data allowed:

- opinion mining
- topic modelling
- discussion trend analysis
- behavioural interpretation

The conversational format of Reddit comments also made the dataset appropriate for the analysis of emotional responses and fan engagement during major football events.

3.7 Dataset Limitations

Several limitations exist within the dataset:

- the data only represents Reddit users and not the broader football audience,
- deleted comments and unavailable accounts reduced completeness,
- sarcasm and football slang may reduce sentiment analysis accuracy,
- the dataset focuses only on a limited tournament period,
- some low-activity users may have limited analytical significance.

Despite these limitations, the dataset remained sufficiently large and diverse to support meaningful social media and network analysis.

4. Methodology

4.1 Overview

This project combines Social Network Analysis (SNA) and Natural Language Processing (NLP) techniques to study football fan conversations in the Reddit community r/soccer during the 2022 FIFA World Cup.

The analytical workflow consisted of:

1. data collection and preprocessing,
2. exploratory data analysis,
3. sentiment analysis,
4. topic modelling,
5. network construction,
6. centrality analysis,
7. community detection,
8. comparative behavioural analysis.

Python was used as the primary implementation language for all stages of analysis.

4.2 Tools and Libraries

Several Python libraries were used throughout the project.

Library	Purpose
pandas	Data manipulation and preprocessing
NumPy	Numerical operations

matplotlib	Data visualisation
seaborn	Statistical visualisation
NetworkX	Graph and network analysis
gensim	Topic modelling
vaderSentiment	Sentiment analysis
scikit-learn	Text preprocessing support
python-louvain	Community detection

These libraries enabled both large-scale text analysis and graph-based analysis within a reproducible Python workflow.

4.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the overall structure and characteristics of the dataset before conducting advanced analysis.

The exploratory analysis included:

- daily posting activity,
- comment activity over time,
- score distributions,
- post flair distributions,
- identification of highly active users,
- identification of highly engaged discussions.

Visualisations were generated to identify patterns and trends within football discussions during the tournament period.

4.4 Text Preprocessing

Text preprocessing was performed before applying NLP techniques.

The preprocessing pipeline included:

- lowercasing text,
- removal of punctuation,

- tokenisation,
- stopword removal,
- removal of invalid or empty text,
- filtering deleted or bot-generated content.

To improve topic modelling performance, we combined post titles and self-text content into one text field.

These pre-processing steps reduced noise and enhanced the quality of downstream NLP analysis.

4.5 Sentiment Analysis

Sentiment analysis was performed using the VADER (Valence Aware Dictionary and sEntiment Reasoner) sentiment analysis model.

VADER was selected because:

- it is specifically designed for social media text,
- it performs well on short conversational messages,
- it captures informal language and emotional intensity effectively.

Each Reddit comment received a compound sentiment score ranging from -1 to +1:

- negative values indicate negative sentiment,
- positive values indicate positive sentiment,
- values near zero indicate neutral sentiment.

Comments were categorised into:

- Positive,
- Neutral,
- Negative.

The sentiment analysis enabled investigation of emotional patterns within football discussions and comparison between user influence and sentiment behaviour.

4.6 Topic Modelling

Topic modeling was performed using Latent Dirichlet Allocation (LDA) with the Gensim library.

LDA was chosen as it is one of the most popular probabilistic topic modelling techniques capable of identifying hidden thematic structures in large collections of text.

The topic modelling process included:

- tokenisation,
- dictionary construction,
- Bag-of-Words corpus generation,
- topic extraction.

The model identified dominant discussion themes across football-related Reddit posts, including:

- match discussions,
- player performances,
- refereeing decisions,
- tournament outcomes,
- fan reactions.

The resulting topics were manually interpreted and labelled based on their most representative keywords.

4.7 Network Construction

A directed user interaction network was constructed using Reddit reply relationships.

Network Definition

- Nodes represent Reddit users.
- Directed edges represent one user replying to another user.
- Edge weights represent the frequency of reply interactions.

The `parent_id` field within Reddit comments was used to identify reply relationships between users.

Self-replies, deleted accounts, and invalid interactions were removed during preprocessing.

The resulting graph enabled analysis of:

- interaction patterns,
- communication structure,
- influential users,
- user communities.

4.8 Centrality Analysis

Several network centrality measures were calculated to identify influential users within the interaction network.

The analysis included:

- in-degree centrality,
- out-degree centrality,
- PageRank centrality,
- betweenness centrality.

PageRank was selected as the primary influence metric because it evaluates both:

- the quantity of interactions,
- the importance of connected users.

This provided a more meaningful measure of user influence than simple interaction counts alone.

4.9 Community Detection

The Louvain algorithm was used for community detection.

The Louvain method is an algorithm that detects densely connected groups of users within a network by maximizing modularity.

We applied the algorithm to undirected version of the interaction graph to find:

- 1.discussion communities
- 2.interaction clusters
- 3.structural separations in the football fan network.

We used the resulting modularity score to assess the strength of community structure in the network.

4.10 Influence and Sentiment Comparison

An additional analysis was performed to examine relationships between:

- 1.user influence
- 2.Popularity
- 3.sentiment behavior.

We combined centrality metrics with sentiment scores to see to what extent more central users showed more positive or negative discussion behaviour.

Such comparative analysis provided further behavioural insights, over isolated NLP or network analysis results.

5. Results

5.1 Exploratory Data Analysis

The dataset consisted of 18,088 posts and 45,020 comments collected from r/soccer between 31 October 2022 and 19 December 2022, covering the entire 2022 FIFA World Cup period. A total of 8,291 unique users authored posts and 12,881 unique users participated in comment discussions, reflecting a broad and active online football community.

Post activity peaked noticeably during key tournament moments. The highest-scoring post during the collection period was “Messi surprised by his mother after winning the World Cup” with a score of 70,417 upvotes and 1,114 comments, followed by the official World Cup final result announcement post with 49,602 upvotes and 9,516 comments. This confirms that emotionally significant moments drove the highest levels of community engagement. The score distribution was heavily right-skewed, with the vast majority of posts receiving modest engagement and a small number of viral posts generating disproportionately high interaction.

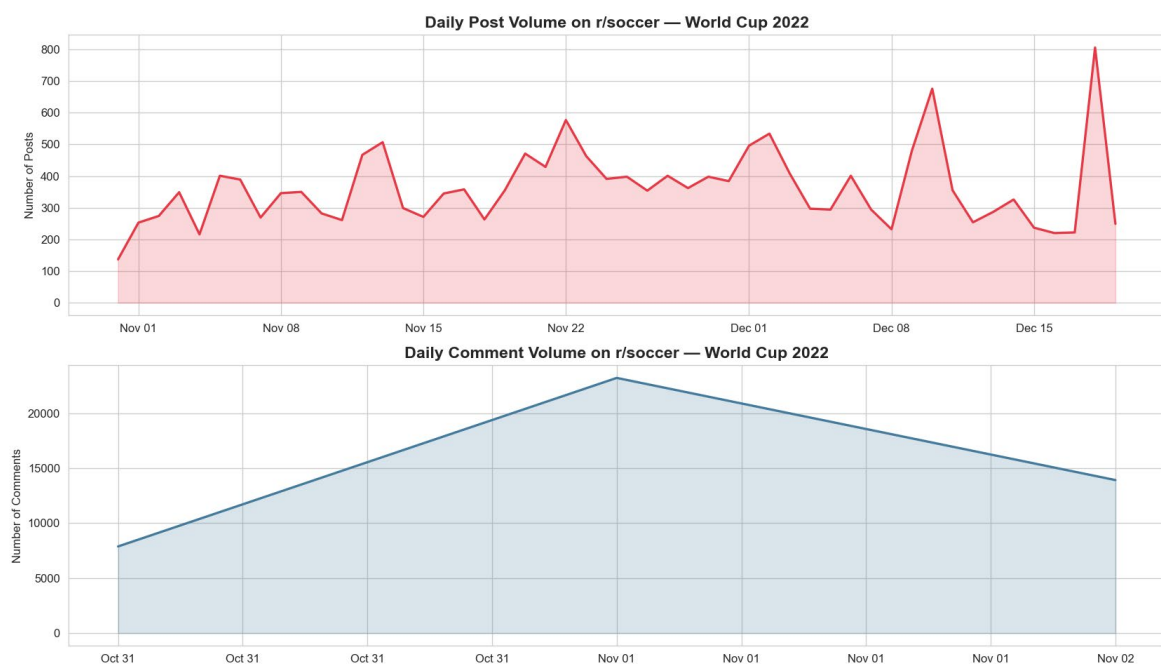


Figure 1. Daily post and comment activity within r/soccer during the 2022 FIFA World Cup period.

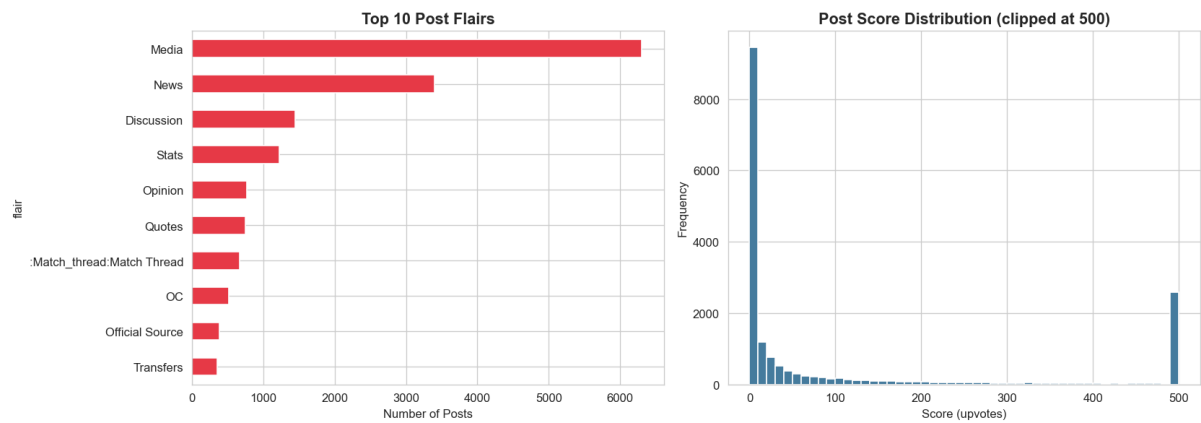


Figure 2. Distribution of Reddit post scores and post flair categories within the dataset.

Post flair categories revealed the dominant content types in the community. Match-related flairs such as “Media”, “Official Source”, and “Goal” (Golaço) were the most frequent, indicating that match reactions and official news drove the majority of discussion activity during the tournament.

5.2 Sentiment Analysis

VADER sentiment analysis was applied to all 45,020 comments. The overall mean compound sentiment score was 0.1116, indicating a mildly positive overall tone across the community during the World Cup. The sentiment distribution broke down as follows: 20,060 positive comments (44.6%), 12,932 neutral comments (28.7%), and 12,028 negative comments (26.7%).

Sentiment trends over time revealed fluctuations corresponding to tournament events. Sentiment dipped noticeably following controversial refereeing decisions and upset results, while positive spikes were observed following iconic moments such as Richarlison’s bicycle kick goal for Brazil against Serbia and Argentina’s World Cup final victory. This temporal pattern confirms that fan sentiment on Reddit is highly reactive to live match events.

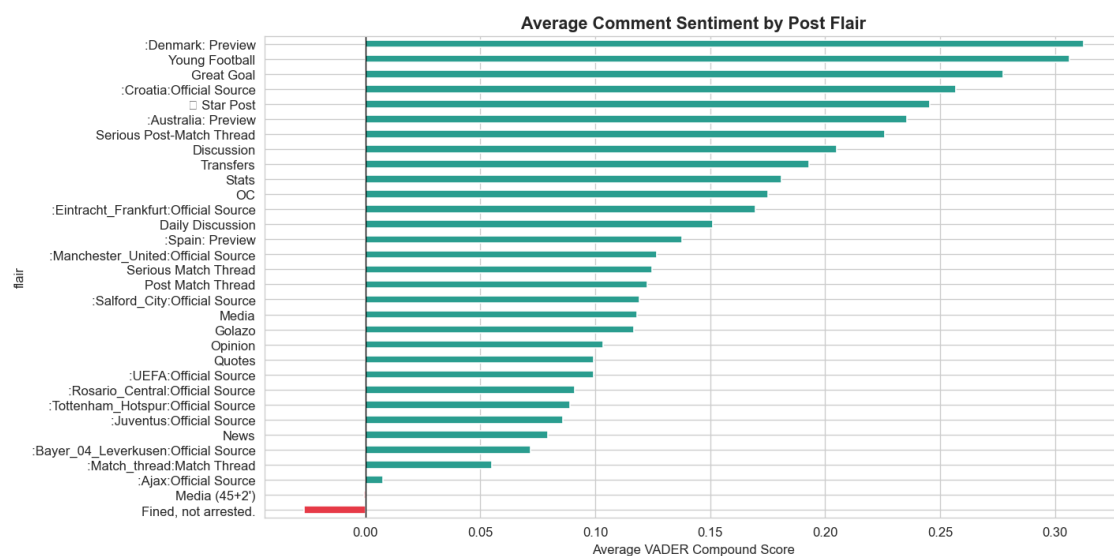


Figure 3. Sentiment distribution and temporal sentiment trends across Reddit comments during the FIFA World Cup.

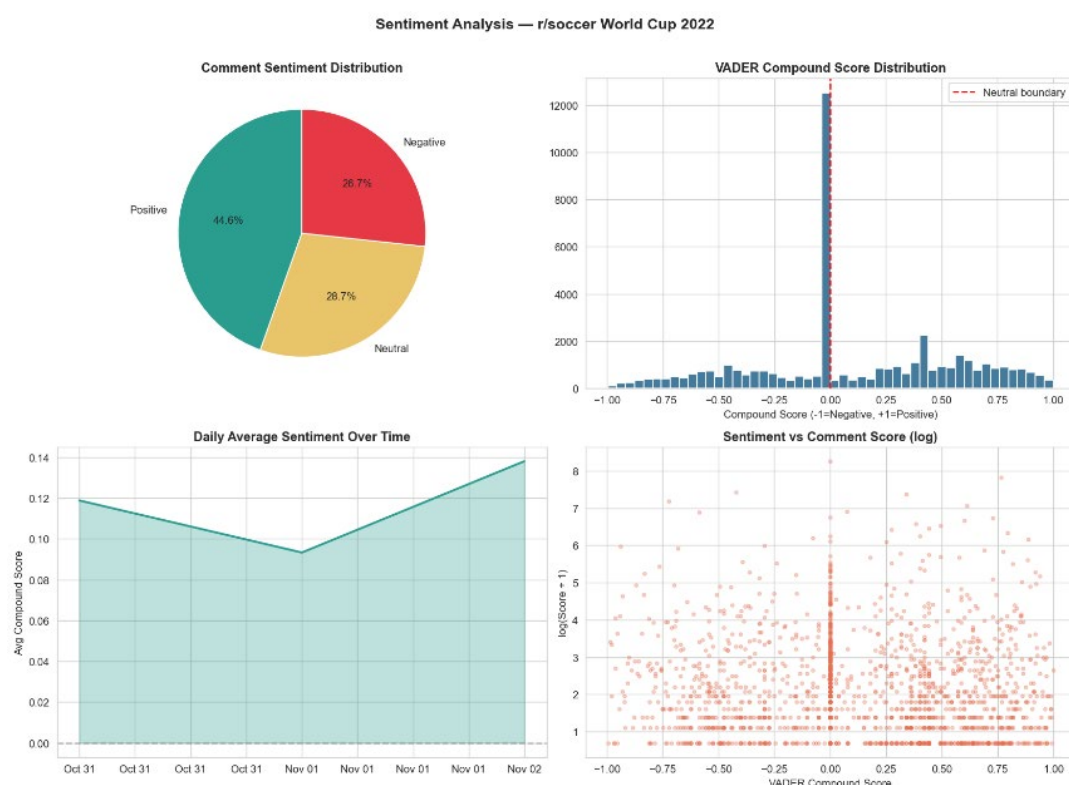


Figure 4. Average sentiment scores across different Reddit post flair categories.

Sentiment analysis by post flair showed variation across content types. Media posts and celebratory goal posts carried higher positive sentiment scores, while posts tagged under controversial topics or refereeing decisions tended towards more negative sentiment. Sentiment was also compared against user influence, explored further in Section 5.4.

5.3 Topic Modelling

Latent Dirichlet Allocation (LDA) was applied to the 18,088 post titles, producing a dictionary of 4,201 terms and a corpus of 16,061 documents after preprocessing. Eight topics were extracted, each manually interpreted based on their highest-weighted terms.

The eight discovered topics and their representative keywords were as follows. Topic 0 (Match Results) was characterised by terms including “final”, “time”, “england”, and “mbappe”. Topic 1 (Tournament Atmosphere) featured terms such as “qatar”, “brazil”, “stadium”, and “neymar”. Topic 2 (Player Performance) was dominated by “ronaldo”, “cristiano”, “manager”, and “fifa”. Topic 3 (Group Stage) included “croatia”, “south”, “korea”, and “stage”. Topic 4 (Argentina/Messi Narrative) was centred on “argentina”, “messi”, “france”, and “winning”. Topic 5 (Fan Reactions) featured “morocco”, “crazy”, and “fun”. Topic 6 (Team Analysis) included “portugal”, “spain”, and “players”. Topic 7 (Club Football News) was characterised by “united”, “real”, “madrid”, and “manchester”.

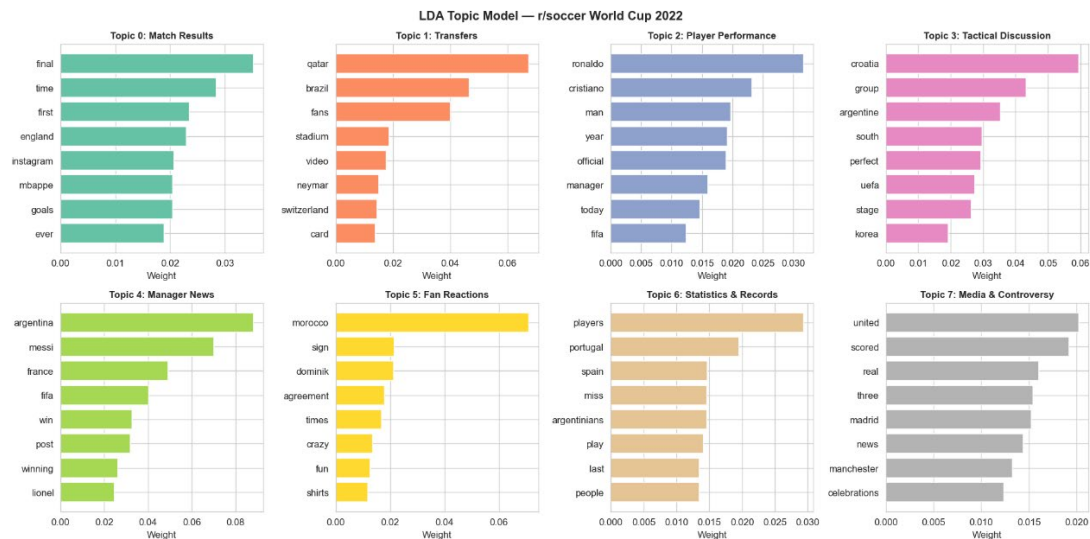


Figure 5. Top representative keywords for each LDA-generated football discussion topic.

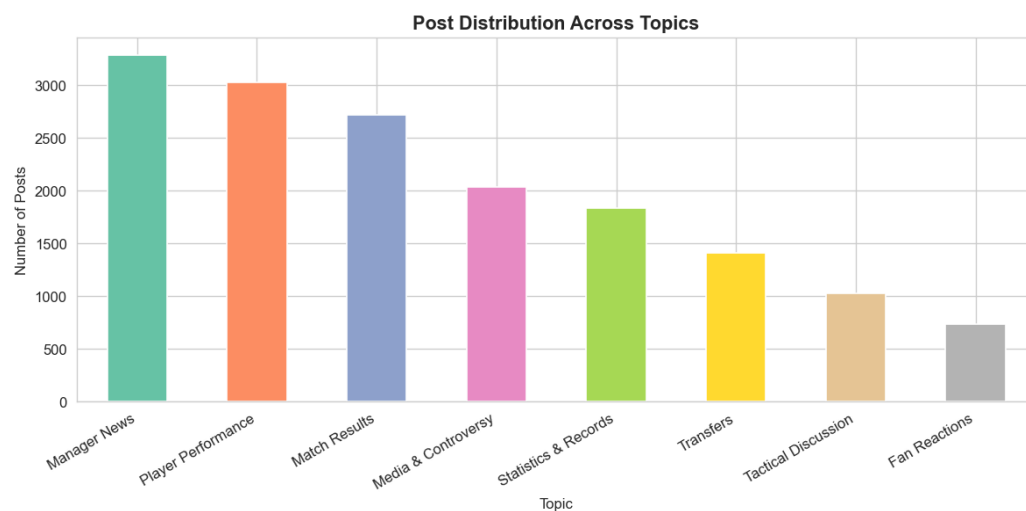


Figure 6. Distribution of Reddit posts across the eight extracted discussion topics.



Figure 7. Word cloud of the most frequent terms appearing in football-related Reddit discussions.

Notably, the Argentina/Messi narrative (Topic 4) and Match Results (Topic 0) dominated the post distribution, reflecting the global fascination with Lionel Messi's World Cup journey and Argentina's eventual victory. Morocco's historic run to the semi-finals was also prominently captured in Topic 5, confirming the significance of unexpected results in driving fan discussion.

5.4 Network Construction and Centrality Analysis

A directed weighted user interaction network was constructed from comment reply relationships. Using the `parent_id` field to identify comment-to-comment replies, 24,862 directed edges were extracted between 10,113 unique users. The largest weakly connected component contained 9,755 nodes and 24,645 edges, with a network density of 0.000259, characteristic of a sparse but well-connected social network typical of large online communities.

Three centrality measures were computed: PageRank, in-degree centrality, and betweenness centrality. PageRank was selected as the primary influence metric as it accounts for both the volume and the quality of incoming connections, rewarding users who receive replies from other well-connected users rather than simply measuring raw interaction counts.

The top-ranked user by PageRank was GreatSpaniard (PageRank: 0.003435, in-degree: 104), indicating this user attracted the highest volume of quality replies within the network. Polskidro ranked second (PageRank: 0.002435, in-degree: 90, out-degree: 133), showing both high influence and high participation as both a content producer and consumer. icemankiller8 ranked third (PageRank: 0.002362, in-degree: 99, out-degree: 106), similarly demonstrating balanced bidirectional engagement.

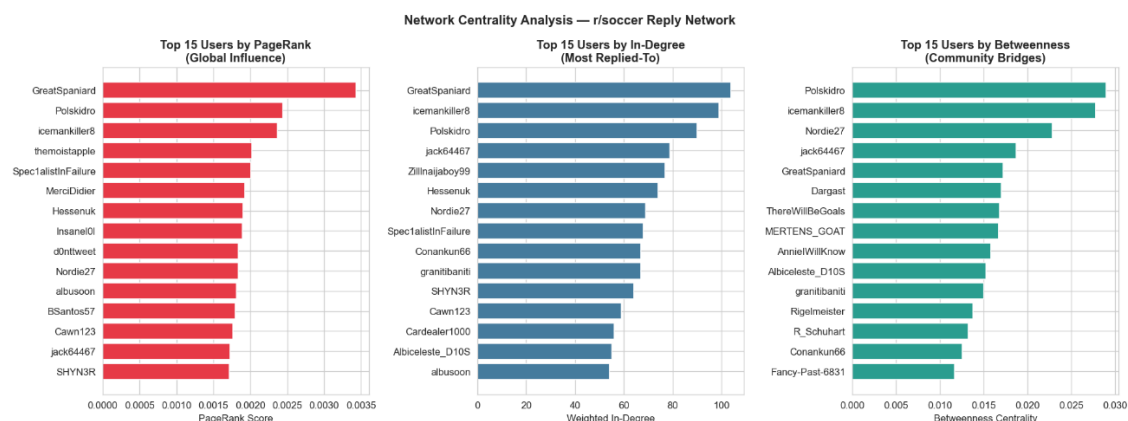


Figure 8. Top users ranked by PageRank, in-degree centrality, and betweenness centrality.

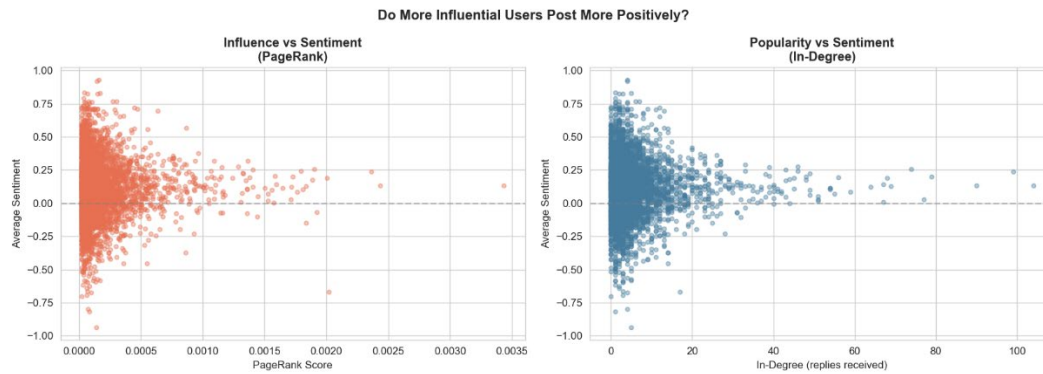


Figure 9. Relationship between user influence metrics and sentiment behaviour.

Betweenness centrality identified users acting as bridges between different discussion clusters. Polskidro had the highest betweenness score (0.028955), suggesting this user connects otherwise separate conversation threads and plays a structurally important bridging role in the network beyond raw reply volume alone. Users with high betweenness but relatively lower PageRank, such as Nordie27 (betweenness: 0.022845, PageRank: 0.001837), represent a distinct category of connector users who facilitate information flow across communities without necessarily dominating any single conversation.

5.5 Community Detection

The Louvain community detection algorithm was applied to the undirected version of the interaction network. The algorithm identified 38 communities with a modularity score of 0.6056. This modularity score is well above the threshold of 0.3 considered indicative of meaningful community structure, confirming that football fans on r/soccer organise into distinct interaction clusters rather than engaging uniformly across the network.

The largest community (Community 3) contained 970 users, followed by Community 6 with 564 users, Community 4 with 544 users, and Community 18 with 541 users. The top 10 communities ranged in size from 333 to 970 users. The presence of both large central communities and many smaller peripheral communities suggests a hierarchical interaction structure, consistent with research on online sports fan communities where discussion naturally clusters around shared team allegiances and topic interests.

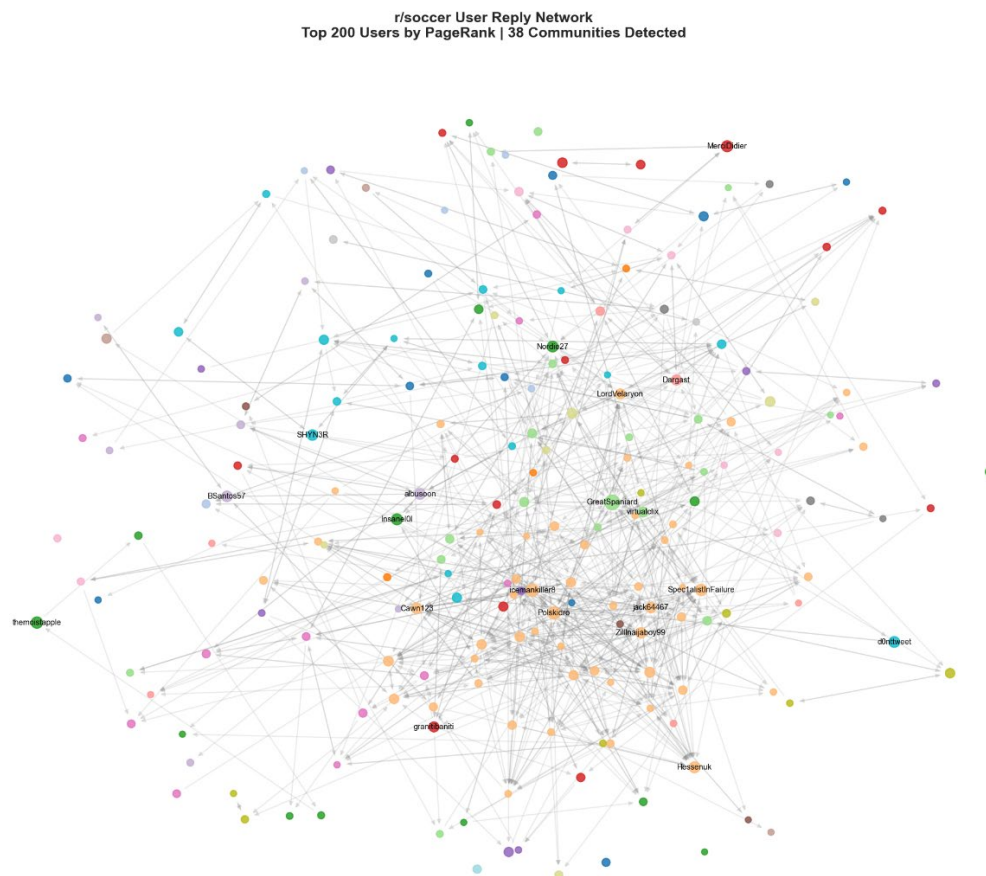


Figure 10. User interaction network visualisation coloured by Louvain community assignment.

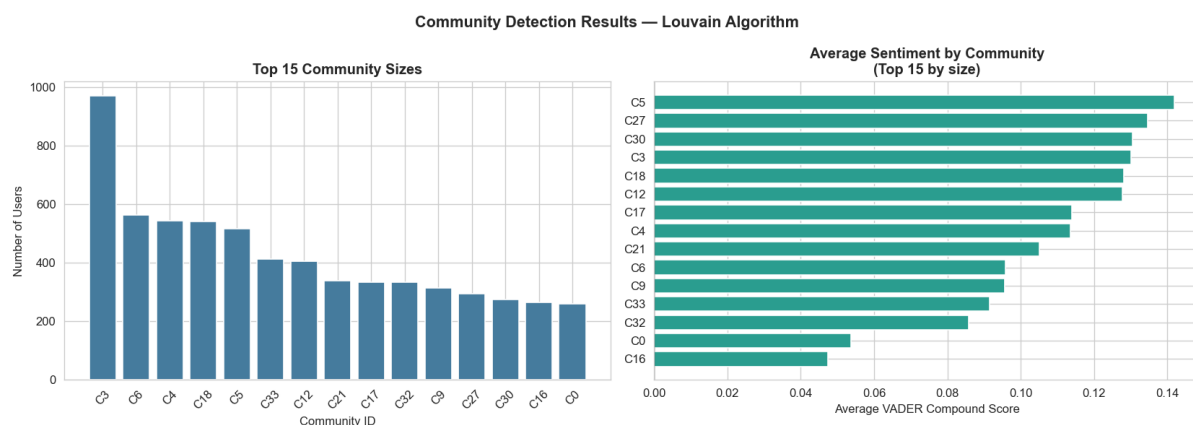


Figure 11. Community size distribution and average sentiment variation across detected communities.

Community sentiment analysis revealed variation in average sentiment across communities. Some communities exhibited consistently positive sentiment, likely centred around celebratory match discussions, while others showed more negative

average sentiment, potentially corresponding to communities focused on controversial refereeing decisions, team eliminations, or critical fan discourse.

6. Discussion

6.1 Answering the Research Question

The research question asked what interaction networks, sentiment polarity, and discussion topics reveal about the social behaviour and influential communities of Reddit football fandom during the 2022 FIFA World Cup. The combined analysis provides several evidence-based answers.

First, the network structure reveals that r/soccer is not a homogeneous community but is instead organised into 38 distinct interaction clusters with strong modularity (0.6056). This suggests fans naturally group around shared interests, team affiliations, or discussion styles, even within a single subreddit. The high modularity score indicates these community boundaries are meaningful and not artefacts of the detection algorithm.

Second, influence within the network is distributed unevenly. A small number of users such as GreatSpaniard, Polskidro, and icemankiller8 attract disproportionately high reply volumes and PageRank scores, consistent with the power-law degree distributions commonly observed in social networks. Importantly, the distinction between PageRank leaders and betweenness leaders reveals two different types of influential users: those who dominate conversations within a community versus those who bridge across communities. This structural distinction has implications for how information and sentiment propagate through the network.

Third, the overall positive sentiment (mean compound score: 0.1116) during the World Cup, combined with temporal fluctuations aligned with match events, confirms that Reddit football communities exhibit collective emotional responses to sporting events. The tournament's emotional arc — from early group stage excitement through Morocco's historic run to Argentina's final victory — is clearly reflected in the sentiment time series.

Fourth, topic modelling confirms that the Argentina/Messi narrative dominated discourse, reflecting the broader global significance of Messi's final World Cup campaign. The presence of Club Football News as a topic (Topic 7) also demonstrates that World Cup discussions are not fully isolated from broader football contexts, with users drawing connections to club football throughout the tournament.

6.2 Limitations

Several limitations should be acknowledged. VADER, while well-suited to social media text, may misclassify football-specific expressions. Terms such as “sick”, “killing it”, or “on fire” carry positive connotations in football contexts but may be scored incorrectly

by a general-purpose lexicon. Sarcasm and irony, common in online football discourse, also present challenges for lexicon-based sentiment analysis.

The dataset is limited to r/soccer and does not represent the broader football fan population. Users on r/soccer tend to be English-speaking and internationally oriented, which may skew topic and sentiment distributions compared to regional or language-specific communities. Deleted accounts and removed comments also introduce incompleteness in the network graph, as reply chains are broken when parent comments are removed.

The network analysis is also limited to the 50,000-comment sample collected. While this sample is large enough to produce statistically meaningful network measures, a complete dataset would likely reveal additional community structures and potentially alter centrality rankings. Finally, community labels were inferred qualitatively from network structure and are not directly observable from the data.

7. Conclusion

This project set out to analyse football fan communities, sentiment, and influential users on Reddit's r/soccer subreddit during the 2022 FIFA World Cup. Using a dataset of 18,088 posts and 45,020 comments, the analysis combined sentiment analysis, topic modelling, network construction, centrality analysis, and community detection to answer the research question.

All six success criteria defined in Section 2 were met. A meaningful and interpretable user interaction network was constructed from 24,862 reply relationships across 10,113 users. Influential users were identified through PageRank and betweenness centrality, revealing both conversation leaders and community bridge users. Thirty-eight distinct fan communities were detected with strong modularity (0.6056), confirming meaningful community structure. Eight coherent discussion topics were extracted via LDA, with the Argentina/Messi narrative and match results dominating discourse. Sentiment trends were successfully extracted and linked to tournament events, with an overall positive community tone (mean: 0.1116). All conclusions are supported by quantitative evidence from the analysis.

The project demonstrates that social media and network analysis techniques can effectively reveal the social dynamics of online sports fan communities. The integration of NLP and graph-based methods provides insights that neither approach could deliver alone, showing not just what fans discuss but how they are structurally organised and who the most influential voices are. Future work could extend this analysis to multiple subreddits for cross-community comparison, apply transformer-based sentiment models to better handle football-specific language, or track how community structure evolves across multiple tournaments over time.

8. References

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